

ASSIGNMENT ON CSE 420

Submitted to : Shazid Bin Mahbab (SZD) sir

Submitted by: Sheikh Md. Tahmid Hossain

ID : 22299522

Section : 12

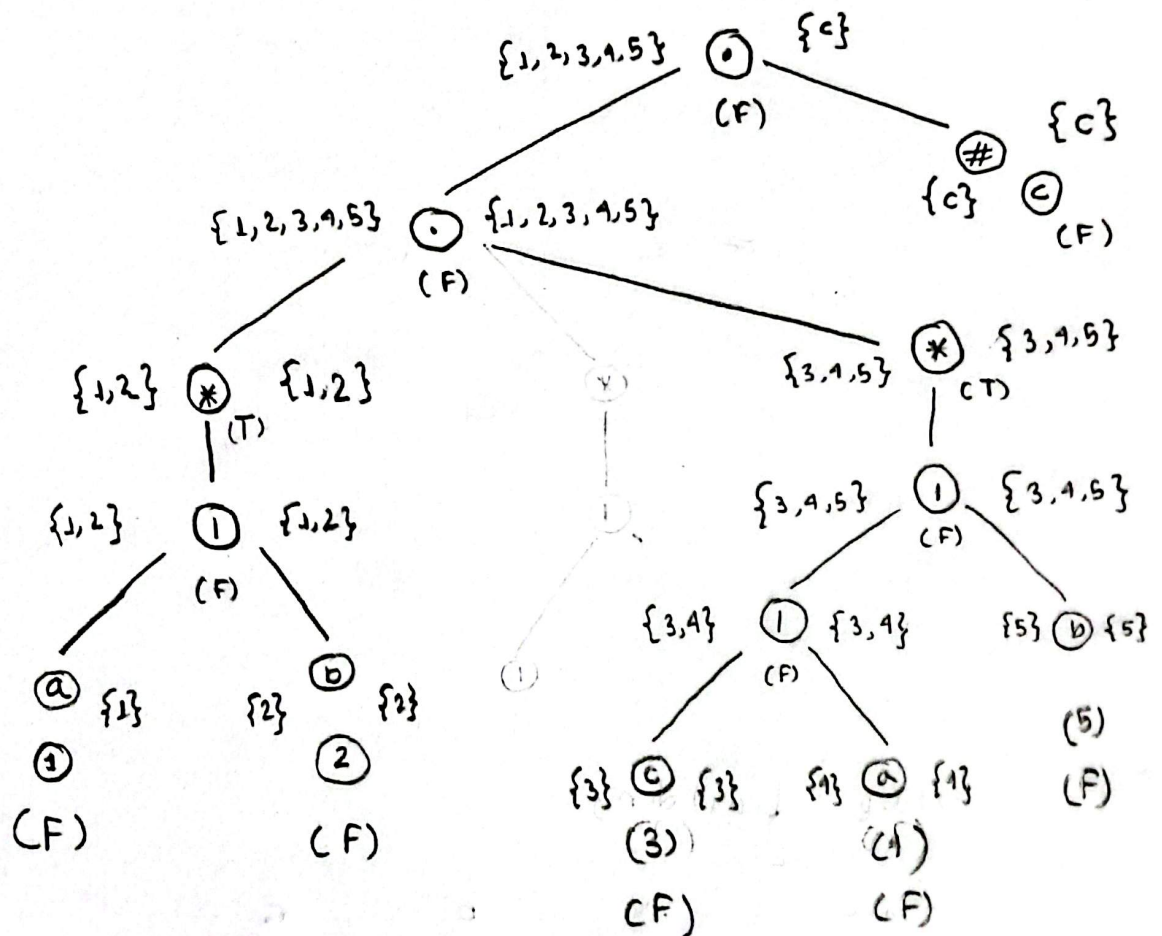
Semester : Summer 26

Assignment No : 1

Date : 12 / 7 / 26

$$RE = (a|b)^* (c|a|b)^*$$

Augmentierte RE = $(a|b)^* (c|a|b)^* \#$



FP Table

Node	FP	Symbol
1	{1, 2, 3, 4, 5, 6}	a
2	{1, 2, 3, 4, 5, 6}	b
3	{3, 4, 5, 6}	c
4	{3, 4, 5, 6}	a
5	{3, 4, 5, 6}	b
6	-	#

State building

State	a	b	c
A	B	B	C
B	B	B	C
C	C	C	C

Start, $\{1, 2, 3, 4, 5\} = A$

$$\begin{aligned} dt(A, a) &= FP(1) \cup FP(4) \\ &= \{1, 2, 3, 4, 5, 6\} \cup \{3, 4, 5, 6\} \\ &= \{1, 2, 3, 4, 5, 6\} = B \end{aligned}$$

$$\begin{aligned} dt(A, b) &= FP(2) \cup FP(5) \\ &= \{1, 2, 3, 4, 5, 6\} \cup \{3, 4, 5, 6\} \\ &= \{1, 2, 3, 4, 5, 6\} = B \end{aligned}$$

$$dt(A, c) = FP(3) = \{3, 4, 5, 6\} = C$$

So, we construct the DFA

